
Understanding where physical theories fail without knowing what replaces them

Chenxi He*

Cavendish Laboratory
University of Cambridge
Cambridge CB3 0HE, United Kingdom

Jingqiu Chen

College of Physics and Optoelectronic Engineering
Shenzhen University
Shenzhen 518060, China

Abstract

Physical predictions can fail before a theory explaining the failure is available. We ask how to test such a failure while restricting the knowledge used to construct the prediction. We make the historically admissible model class an explicit input to a calibrated test: a calendar date, the knowledge horizon, fixes which physical laws may construct the prediction, and later theories enter only for retrospective comparison. Measurements are compared with a named prediction under an observation model, and generic screens rank relations by how unlike the admissible laws they are. At an 1899 horizon, all five pre-1900 predictions are rejected on five selected historical datasets. On the same measurement grids and error models, four of the five historical successors are not rejected, whereas retrospectively chosen alternatives with the wrong shape or mechanism remain rejected. Millikan’s nearly perfect photoelectric line rejects the classical prediction: curve familiarity is not physical compatibility. A pre-1900 representation ranks later law families above pre-1900 controls, with performance that depends on how the reference formulas were collected and generated. The historical constraint thus allows the predictive domain of specified physical models to be tested without using a successor theory to construct the test.

1 Introduction

The failure of a physical prediction can be evident before its explanation is known. Classical predictions for radiation, heat capacity and electrical resistance provide familiar examples: measurements outside a successful regime eventually required quantum descriptions. A test of predictive failure, however, requires a reference prediction and an account of the measurement errors. The question is how to construct that reference while excluding the knowledge that later explained the departure.

Existing methods quantify discrepancy, detect change or search for replacement laws. Bayesian model discrepancy compares observations with a simulator [1], conformal prediction calibrates residuals [2], change-point methods locate distributional shifts [3] and out-of-distribution detectors score departures from a reference distribution [4]. Machine-learning methods interpolate, forecast or extrapolate scientific data [5, 6], and symbolic

*Correspondence: ch2067@cam.ac.uk

31 regression searches for governing expressions [7, 8]. Each takes the reference model or the
32 space of candidate laws as given. None specifies which physical models were admissible
33 before the later explanation existed.

34 We make that specification an explicit input. A calendar date, the knowledge horizon, fixes
35 the knowledge a test may use: a relation is admissible if its physical content follows from laws
36 and principles documented before the horizon. The pre-horizon prediction for a measured
37 quantity is built from admissible relations only. The historical successor that later explained
38 a departure enters afterwards, for comparison on the same measurements, so it never defines
39 the discrepancy being tested.

40 EPOCH (Epistemic Probe of Classical Horizons) separates generic screens from tests of
41 named predictions. A representation screen measures how far a curve lies from pre-horizon
42 relations in the embedding of a neural encoder trained only on those relations. A simple-form
43 extrapolation screen, which fits power, exponential, linear or constant forms, fits the best form
44 over the stretch of data it describes, the fitted regime, and scores its extrapolation over the
45 remainder, the extrapolation region. Both screens identify unfamiliar curves; neither refers
46 to a physical theory. A named-prediction test compares the residuals of the measurements
47 from a specified pre-horizon prediction with a null distribution simulated under the stated
48 measurement errors. It returns one of four verdicts: compatibility over the measured range;
49 localized incompatibility, when the prediction fits a regime and fails beyond an estimated
50 boundary; whole-range incompatibility, when the prediction is rejected over the sampled
51 range and no compatible regime is identified; or abstention, when the inputs do not support
52 a test.

53 We use an 1899 horizon, for which classical relations can be reconstructed from dated sources.
54 We first show how the horizon defines the admissible prediction for a dataset. We then test
55 five historical predictions on measurements and replace each by its successor on the same
56 grid and error model. Next we ask how far the generic screens rank later laws above pre-1900
57 laws and where they stop being informative. Finally we show that the observation model
58 determines both rejection and the localization of a boundary.

59 2 Results

60 2.1 A historical cutoff defines the admissible prediction

61 Figure 1a places dated law families on either side of the 1899 horizon. Relations sharing a
62 law form a law family, and training and evaluation are divided by family. Pre-1900 families
63 supply both the relations on which the representation screen is trained and the predictions a
64 test may use; later families enter only for evaluation. A test (Fig. 1b) compares measurements
65 with the admissible pre-1900 prediction under an observation model, the stated description of
66 the errors, censoring and grid, and returns a verdict. The generic screens form a side branch
67 that flags relations for testing (Fig. 1b,c). They receive only a point cloud, the normalized
68 (x, y) pairs of a relation without formula, units or labels. Where a fitted regime is sampled,
69 the extrapolation screen can also localize a departure, which gives a descriptive classification;
70 rejection of a named physical prediction comes from the named-prediction test. A fixed
71 rule assigns each dataset to the tests its inputs support (Methods; Supplementary Fig. 1).
72 Figure 1d shows the verdicts on real data, and Table 1 summarizes the evidence.

73 2.2 Historical predictions fail where their successors fit

74 All five pre-1900 predictions are rejected, and on the same grids and error models four of the
75 five successor predictions are not rejected (Fig. 2). We selected these measurements as known
76 classical failures. The COBE-FIRAS spectrum and Bertozzi’s electron data lack a measured
77 classical regime, whereas five heat-capacity curves and mercury resistance contain one, so
78 the cases distinguish failure across the sampled range from failure beyond a fitted regime
79 (Fig. 2a–d). Millikan’s photoelectric line tests a different distinction: a familiar mathematical
80 shape can violate a named physical prediction.

81 The COBE-FIRAS monopole spectrum of the cosmic microwave background [9] has 43 points
82 and, at $T = 2.725$ K, samples $x = h\nu/kT$ from 1.2 to 11.3. The complete range lies outside

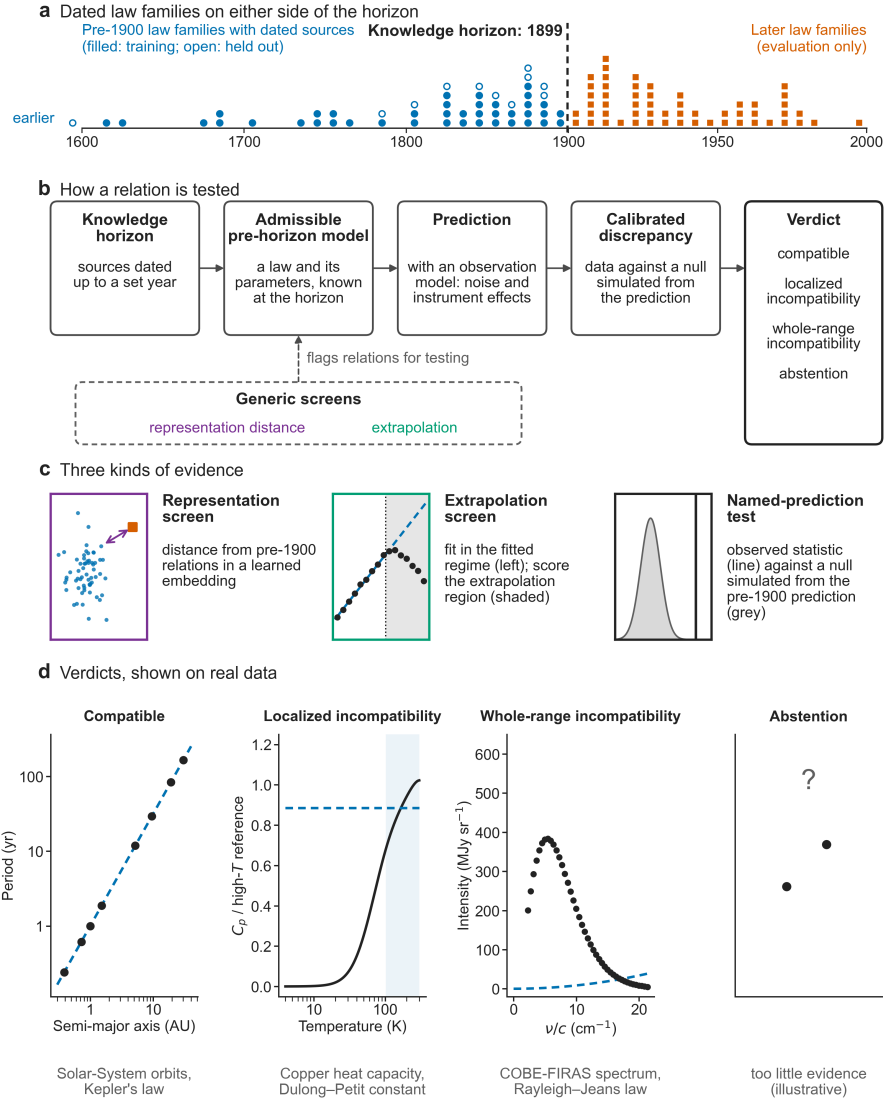


Figure 1. Testing a prediction under a knowledge horizon. **a**, Pre-1900 law families with dated sources (blue circles; filled, used to train the compact encoder; open, held out) and the later law families of the benchmark (vermillion squares), one point per family at its first dated source. The time axis changes scale at 1900. The open circle marked ‘earlier’ is a held-out family dated about 250 BC. **b**, How a relation is tested. The knowledge horizon (sources dated up to a set year) fixes the admissible pre-horizon model, a law and its parameters known at the horizon; its prediction, with an observation model for noise and instrument effects, gives a discrepancy calibrated against a null simulated from the prediction, and the test returns one of four verdicts. The generic screens (dashed box; representation distance and extrapolation) form a side branch that flags relations for testing; where a fitted regime is sampled, extrapolation can also localize a departure. **c**, Schematic of the three kinds of evidence: the distance of a new point cloud (normalized (x, y) pairs without formula or units; vermillion square) from pre-1900 relations in the encoder’s embedding (representation screen); a form fitted in the fitted regime (solid blue) and extrapolated to the extrapolation region (dashed, grey shading), with measurements in black (extrapolation screen); and an observed statistic (black line) against a null distribution simulated from the pre-1900 prediction (grey) (named-prediction test). **d**, Verdicts on real data, with pre-1900 predictions dashed: Solar-System orbits under Kepler’s law, a classical control (compatible); the National Institute of Standards and Technology (NIST) copper heat-capacity curve under the Dulong–Petit constant fitted in the shaded fitted regime (candidate localized incompatibility; normalized as in Fig. 2c); the Cosmic Background Explorer Far Infrared Absolute Spectrophotometer (COBE-FIRAS) spectrum under the fitted Rayleigh–Jeans law (whole-range incompatibility). The abstention panel is illustrative.

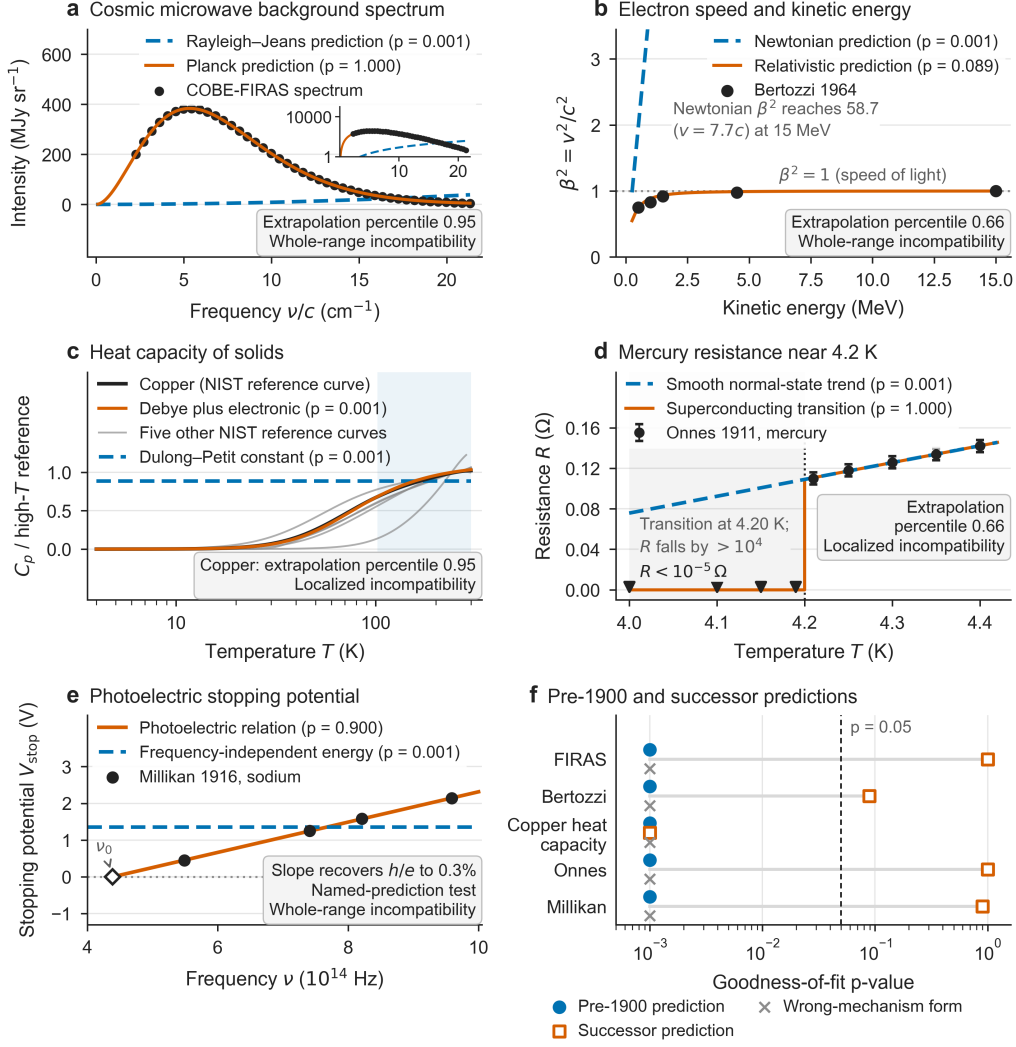


Figure 2. Historical prediction tests under the 1899 horizon. **a–d**, Black points are published processed measurements or digitized historical values; in **c** the black curve is the NIST reference curve for copper and the grey curves are the five other solids. Blue dashed curves are the pre-1900 predictions as fitted in the goodness-of-fit test (for mercury, the smooth normal-state trend; for copper, the constant fitted over the shaded fitted regime) and vermilion curves the successor predictions fitted on the same grid (Planck, relativistic kinetic energy, Debye plus electronic term and a superconducting transition); legends give the goodness-of-fit p -value of each named prediction (also in **f**), not a test of boundary localization. Grey boxes give the extrapolation-screen percentile, the empirical percentile of the extrapolation score among the size-matched classical calibration scores (upper-tail p -values in Supplementary Note 4), and the verdict; the localized verdicts in **c** and **d** are descriptive candidate supplementations, and in **e** the box gives the slope check and the test giving the verdict. The inset in **a** repeats the spectrum on a logarithmic intensity scale. FIRAS and Bertozzi yield whole-range incompatibility. Specific heat and Onnes contain a fitted regime and a candidate boundary. In **c** heat capacity is normalized by its median above 70% of the maximum temperature. Shading marks the fitted regime in **c** and the region beyond the transition in **d**; in **d** the dotted line (labelled 4.20 K) marks the historically recorded transition temperature, error bars give the figure-reading uncertainty and downward triangles mark the four values below the reported limit of $10^{-5} \Omega$. **e**, In Millikan’s 1916 digitized values, stopping potential rises linearly with frequency as $V_{\text{stop}} = (h/e)(\nu - \nu_0)$. The photoelectric relation (vermilion) recovers h/e to 0.3%, and the open diamond marks the extrapolated zero crossing ν_0 . The extrapolation score falls below the threshold, whereas the named-prediction test rejects the classical prediction of frequency-independent energy (blue dashed line). **f**, Goodness-of-fit p -values on the same measurement grid under the pre-1900 prediction (filled blue circles), its successor (open vermilion squares) and a wrong-mechanism form (grey crosses), joined by a grey line for each dataset; the vertical dashed line marks $p = 0.05$.

Table 1. Summary of the evidence. Each row gives one headline result. AUROC (0.5 is chance) and the same-generator set are defined in the text. Intervals are 95% bootstrap intervals over families or mechanism groups. Every quantitative comparison and its evaluation population is listed in Supplementary Note 19.

Evidence	Headline result	Interpretation
Historical prediction tests, five selected datasets	All five pre-1900 predictions rejected; four of five successors not rejected on the same grid and error model	A stated prediction can be tested without using its successor to construct the test
Generic-screen ranking, generated formulas	Large encoder ranks 55 later families above nine pre-1900 control groups with AUROC 0.925 [0.851, 0.982], and the same-generator set at 0.536 [0.424, 0.643]	Ranking of later relations depends on the formula corpus and its generator
Measured transfer, sixteen series in 5 against 9 groups	Compact encoder 0.62 [0.29, 0.89]; extrapolation screen 0.78 [0.38, 1.00]; both intervals span chance	Transfer to measurements not demonstrated
Detection and observation effects, synthetic	True-positive rate 0.271 [0.073, 0.500] at the 5%-targeted threshold (0/96 false positives); false-positive rates 0.40 under saturation and 0.63 under censoring at the highest simulated severity	Calibrated detection is weaker than ranking; departures need an observation model

$x \ll 1$, where the Rayleigh–Jeans law [10, 11] holds. Its prediction, intensity proportional to ν^2 , over-predicts the intensity above the peak by orders of magnitude, by an amount that depends on the fitting window (Supplementary Note 5). The verdict is whole-range incompatibility; the data do not locate a transition from a classical regime.

Bertozzi’s electron measurements [12] likewise lie outside the non-relativistic regime. For kinetic energy against $\beta^2 = v^2/c^2$, the classical law $\frac{1}{2}mv^2$ crosses $\beta^2 = 1$ at 0.26 MeV and reaches $\beta^2 = 58.7$ at 15 MeV, equivalent to $7.7c$. The lowest measured point is already $0.87c$, and the measured β^2 values approach 1. The verdict is whole-range incompatibility with the Newtonian energy–velocity prediction.

The Dulong–Petit law [13] predicts temperature-independent molar heat capacity. Five of the six NIST cryogenic reference curves contain a high-temperature plateau, the fitted regime, and depart from that constant at low temperature, giving a candidate localized incompatibility; beryllium lacks a plateau and gives whole-range incompatibility. For oxygen-free high-conductivity copper, the extrapolated plateau over-predicts the 4 K reference value by 3.7×10^3 . The six reference curves share a low-temperature mechanism and are grouped together.

Kamerlingh Onnes’s mercury resistance values [14] contain a smooth normal-state trend followed by a discontinuous drop. The transition is placed at 4.20 K, a temperature fixed from the historical record in the paired prediction test. The smooth normal-state and residual-resistance models [15] are tested with figure-reading uncertainty and the reported resistance bound below $10^{-5} \Omega$ in the observation model.

Millikan’s 1916 measurement [16] demonstrates that curve familiarity is not physical compatibility. Stopping potential is linear in frequency, $V_{\text{stop}} = (h/e)(\nu - \nu_0)$, with $R^2 = 0.99995$ and a slope recovering h/e to 0.3% (Fig. 2e). The extrapolation screen accepts the shape, yet the named-prediction test on the same four measurements rejects the classical prediction of frequency-independent photoelectron energy. The fitted intercept gives $\nu_0 = 4.39 \times 10^{14}$ Hz; all four frequencies exceed ν_0 , so the verdict is whole-range incompatibility.

Each pre-1900 prediction was then replaced by its historical successor and scored on identical measurements and grids under the same error model (Fig. 2f); successor nuisance parameters are fitted to the same data where specified, and the Onnes successor is a phenomenological transition form. All pre-1900 predictions are rejected (bootstrap $p = 0.001$). The successors are not rejected for FIRAS under Planck’s law [17], Bertozzi under relativistic kinetic

energy [18], Onnes under a superconducting transition model and Millikan under the photoelectric relation [19]; individual p -values are given in Fig. 2 and Supplementary Note 11. The copper successor, a Debye-plus-electronic form [20], is rejected at the primary 5% reference-curve error floor ($p = 0.001$) and passes when the floor is raised to 7.5%; the floor stands in for a covariance model of the reference curve, which is unavailable.

Wrong-shape and wrong-mechanism alternatives, one of each per case and retrospectively chosen after the datasets were known, remain rejected: all five wrong-shape alternatives after Benjamini–Hochberg correction ($q \leq 0.0015$) and all five dimensionally coherent wrong-mechanism alternatives (1,000 bootstrap replicates; $p = q = 1/1001$). The tested alternative forms therefore do not reproduce the successors’ agreement (Supplementary Note 11).

Measured classical controls retain their expected forms: Solar-System orbits [21] and the four Galilean moons recover Kepler’s exponent of $3/2$ to four figures (1.500 and 1.500), and Boyle’s 1662 pressure–volume table [22] gives -0.998 ; all three fall below the extrapolation screen’s 0.5 screening threshold (Supplementary Fig. 2a–c). The extrapolation screen does not reject FIRAS, copper heat capacity, Bertozzi or Onnes after Benjamini–Hochberg correction (Supplementary Note 4), so the rejections come from the named-prediction tests, and the localized verdicts for heat capacity and resistance are descriptive classifications.

2.3 Generic screens rank later laws but do not replace a test of a physical prediction

A pre-1900 representation ranks later laws, with performance that depends on how the reference formulas were collected and generated. The large encoder, trained only on point clouds from dimensionless reductions of pre-1900 laws, ranks 55 later law families above nine held-out pre-1900 control groups with AUROC 0.925 [0.851, 0.982] (Fig. 3a). AUROC, the area under the receiver operating characteristic curve, is the probability that a randomly chosen later family outranks a pre-1900 group, counting ties as one half; 0.5 is chance. Brackets give 95% bootstrap intervals over families or groups. The untrained network, the same network with its initial random weights, gives 0.649 on the same comparison, and each of the encoder’s four distance components improves on it (Supplementary Fig. 4d).

The ranking depends on how the reference formulas were collected and generated. Regenerating the nine control groups with the later families’ sampling and noise still gives AUROC 0.857 [0.741, 0.945], against 0.545 for the untrained network (Fig. 3a); these regenerated controls were added after the primary results were known. The dependence appears on the same-generator set, a separate collection of later formulas dated 1901–1950 and produced by the generator of the training corpus, whose family composition also differs from the 55-family benchmark: there the trained encoder gives 0.536 [0.424, 0.643], near chance. The learned ranking therefore reflects which formulas were collected and how they were generated, as well as when the laws were found. An exploratory extension to a 1950 horizon, whose added formulas lack complete source-by-source dates, gives 0.453 [0.228, 0.678] on post-1950 families (Supplementary Note 16).

Simple-form extrapolation also ranks the later families above pre-1900 controls, without a trained network. Against eleven cited pre-1900 families excluded from encoder training, the extrapolation screen gives AUROC 0.888 [0.800, 0.959], compared with 0.669 [0.534, 0.798] for a compact encoder trained on 205 point clouds from 40 source-verified pre-1900 families (Fig. 3b). The compact encoder ranks 12 synthetic departure families, curves built to leave a classical form, above 12 classical families with family AUROC 0.903, against 0.375 for an encoder matched in architecture, training and data size but trained on generic smooth expressions (Supplementary Fig. 4c; Supplementary Note 8). Ranking is stronger than calibrated detection: at a threshold targeted to 5% false positives, the extrapolation screen detects synthetic departures with true-positive rate 0.271 [0.073, 0.500] (Supplementary Note 4), and at their calibrated thresholds each screen flags 7 of the 55 later families and none of the 11 controls (Supplementary Note 15).

The families the screens miss show why a ranking does not test a physical prediction. Threshold and piecewise relations separate more strongly by extrapolation, whereas affine, power and exponential relations can remain close to the admitted forms. Later laws with familiar shapes, such as special relativity, the photoelectric threshold and radioactive decay,

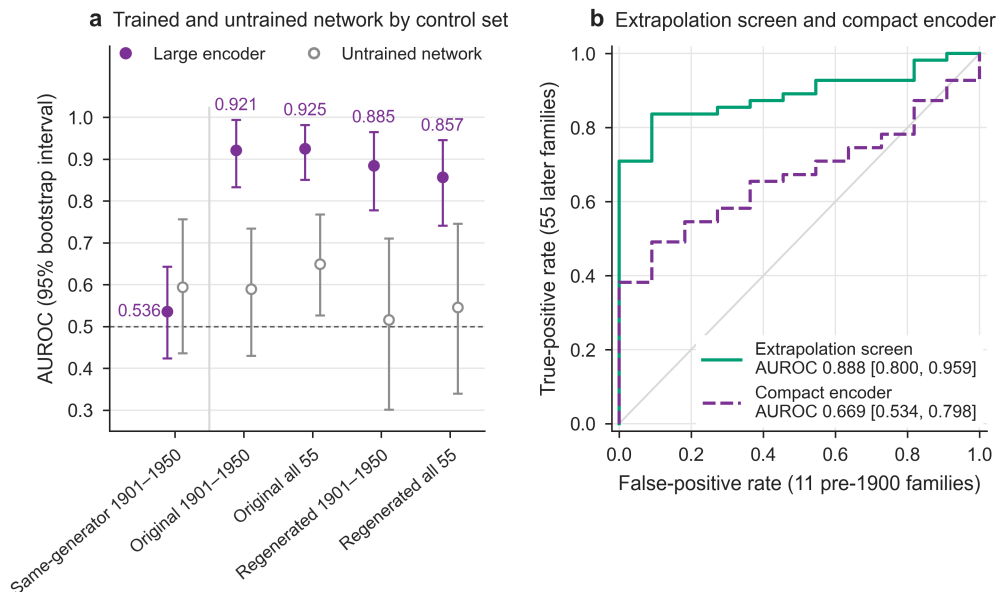


Figure 3. Generic screening at the 1899 horizon. **a**, AUROCs of the large encoder (filled purple, values labelled) and the untrained network (open grey) for the same-generator set and for each combination of later families (1901–1950 or all 55) and pre-1900 controls (original or regenerated), with 95% bootstrap intervals over families or groups; the same-generator comparison is shown left of the vertical line and the dashed line marks chance. **b**, Receiver operating characteristic curves of the extrapolation screen (green) and the compact encoder (purple, dashed) for the 55 later families against 11 cited pre-1900 families excluded from encoder training; the diagonal marks chance. Learned score distributions, the compact encoder’s controls and the large encoder’s ROC curves and distance components are shown in Supplementary Fig. 4.

are flagged by neither screen and require a named-prediction test (Supplementary Note 15). Eleven further generated cases, assembled after the primary scores were known, illustrate the same distinction (Supplementary Fig. 5; Supplementary Note 17). On sixteen measured NIST series, grouped into five post-1900 and nine classical or metrological mechanism groups and collected during method development, the compact encoder gives AUROC 0.62 [0.29, 0.89] and the extrapolation screen 0.78 [0.38, 1.00]; both intervals span chance (Supplementary Fig. 6; Supplementary Note 18).

2.4 Observation models determine rejection and localization

The observation process can produce departures even when the underlying classical relation is unchanged. At the highest simulated severity, false-positive rates of the extrapolation screen are 40% for saturation and 63% for censoring under calibration targeted to 5%; the other tested effects remain at or below 11% (Fig. 4a). A plateau or threshold therefore requires an observation model that accounts for these effects, as the dependence of the copper comparison on its error floor also shows.

Localizing a boundary requires intervals calibrated beyond ordinary measurement resampling. When a compatible regime and its departure are both sampled, a nominal 90% measurement-bootstrap interval covers 53.9% of simulated boundaries (358 simulations), and 8.5% for steps. We calibrated the interval radii on 899 independent simulations, separately by sample size, raising held-out coverage to 91.3% [89.3%, 93.0%] across 900 simulations (Wilson interval; Fig. 4b). This is marginal coverage under the specified generators (Supplementary Note 13).

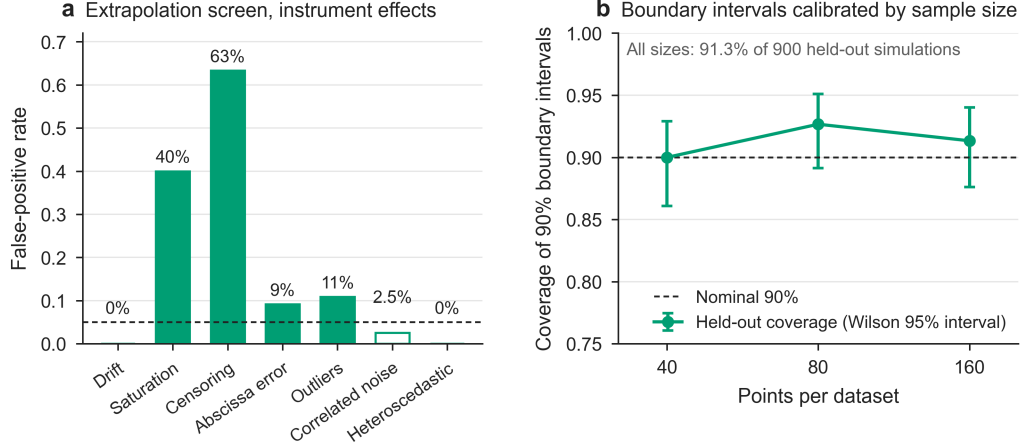


Figure 4. Observation effects and boundary calibration. **a**, False-positive rates of the extrapolation screen under simulated instrument effects (abscissa error: noise in the independent variable) at the highest severity tested, with calibration targeted to 5%; the dashed line marks the nominal rate, open bars are at or below it and rates of 0% have no visible bar. **b**, Held-out coverage of 90% boundary intervals calibrated separately at each sample size (40, 80 and 160 points); points give coverage, bars give Wilson 95% intervals and the dashed line marks nominal coverage.

3 Discussion

Detecting an unusual curve is not the same as testing a physical prediction. A screen ranks a curve against a reference corpus; only a named prediction, stated with its observation model, can be rejected. A historical knowledge constraint makes that test possible for predictions whose failure was later explained, without letting the successor define the discrepancy: the reference is built from pre-horizon physics, and the successor enters afterwards on the same grid and error model. In four of the five historical comparisons this replacement turns rejection into non-rejection, whereas the tested alternative forms keep their discrepancies.

Millikan’s line and a classical interference resonance show why the distinction matters. Millikan’s line has an ordinary shape, yet its frequency dependence rejects the classical prediction (Fig. 2e). Conversely, a sharp interference resonance in a measured transmittance series follows from classical wave optics (Supplementary Fig. 7). Extrapolation of a fitted generic form tests a shape, and the representation screen measures familiarity; a claim about a theory needs that theory’s prediction.

EPOCH combines neural representation methods [5, 6], calibrated residuals and model-discrepancy testing [2, 3, 1]. It is related to model-independent searches in particle physics [23] and to machine rediscovery of laws [7, 8, 24]. Symbolic regression can seek candidate replacements under a parsimony prior [25]; testing a restricted prediction and searching for its replacement address different questions. The historical constraint specifies which knowledge can enter the reference prediction.

The evidence has limits. Because the measurements were selected as known failures, the historical comparisons demonstrate the construction but do not give a prospective detection rate. Rejection and localization depend on the observation model, as copper’s error-floor sensitivity and the saturation and censoring simulations show. The learned ranking depends on the formula corpus and its generator, and transfer of the screens to measured series and to a later horizon has not been demonstrated. On measured data the largest learned scores belong to a standard-resistor record and a polymer-impedance curve, so a high learned score marks an unusual relation, not a physical mechanism.

A prospective study would pair independently selected measurements with a dated corpus, specified pre-horizon predictions and externally supported error models. With a simulator-based reference, the same specification extends to physical descriptions that supply predictions

without a closed-form law, and screening can direct attention to relations that a named-
prediction test then examines. Making the admissible knowledge explicit allows the predictive
domain of a historical model class to be tested without the theory that replaced it.

4 Methods

4.1 Historical admissibility

A relation is admissible at the knowledge horizon, here 1899, if its physical content follows
from laws and principles documented on or before that date. The date is set by the source
law, not by when a particular formula text was printed. Mathematical rewriting that adds
no physical content, such as nondimensionalization, reduction to one variable or algebraic
rearrangement, is allowed. The Rayleigh–Jeans form follows from the equipartition theorem
applied to the electromagnetic modes of a cavity, and the normal-state resistance model from
Matthiessen’s 1864 decomposition of resistance into a temperature-dependent part and a
constant residual part [26]; both are therefore admissible pre-1900 predictions. Where the
standard statement of a prediction was published later, as for the Rayleigh–Jeans form in
1900 and 1905 and the Drude model in 1900, we cite that statement as the reference and do
not use it as a training source. Only admissible relations enter the training corpora and the
predictions a test may fit.

Relations derived from one physical law traced to one dated source form a law family, and
measured series produced by one physical mechanism form a mechanism group. Both are
the units over which AUROC are computed and bootstrap intervals resampled; the nine
pre-1900 control groups are sets of law families that share one mechanism.

4.2 Specifying a test

An EPOCH test specifies $\mathcal{T} = (\mathcal{M}, \Theta, P_\varepsilon, I, A, S, \alpha)$. Here $\mathcal{M}$ lists the admissible models,
 Θ the fitted parameters, P_ε the observation model, I the fitted regime, A every adaptive
choice, S the discrepancy statistic and α the error rate. The observation model comprises
the error distribution, censoring or saturation, and the grid; where no published covariance
exists, an error floor assigns a minimum relative error. Direction choice, window search,
model selection and parameter fitting all belong to A and are rerun inside every calibration
and null replicate.

The named-prediction test computes the residuals of the measurements from the pre-horizon
prediction, scaled by the stated errors, and compares their size with a null distribution:
the distribution of the statistic over datasets simulated from that prediction on the same
 x -grid with the same error model. Historical named-prediction p -values use parametric
bootstrap simulations under the specified observation model, with nuisance parameters fitted
to the data and refitted in each replicate; their calibration depends on that model and
on the bootstrap approximation. Calibrated screening tests use the upper-tail probability
 $p = (1 + \#\{S_i \geq S_*\})/(n_{\text{cal}} + 1)$ with ties counted conservatively, and dataset scans report
Benjamini–Hochberg adjusted q -values [27]. For exchangeable calibration and test units, this
rank-based construction has finite-sample validity. AUROC intervals resample law families
or mechanism groups. In the synthetic calibrated-detection evaluation of the extrapolation
screen, the primary reported quantity is the true-positive rate at a threshold targeted to
5% false positives on independent calibration, with the 1% threshold secondary; screening
comparisons report AUROC. The 0.5 threshold on the extrapolation-screen percentile (the
calibration median; Fig. 2) selects cases for further testing.

The assignment rule (Supplementary Fig. 1) sends a dataset to the extrapolation screen,
whose classification is descriptive, when an admitted form fits a contiguous fitted regime,
otherwise to the named-prediction test when a pre-horizon prediction and error model are
supplied, and otherwise to abstention; when a fitted regime exists and a pre-horizon prediction
and error model are also supplied, both tests are run and reported, as for Millikan’s data.
Nearly constant responses go to the named-prediction test, which tests a constant against
noise, and point clouds with at least 20 observations also receive a learned score.

272 Specification timing differed across analyses. The historical comparisons are retrospective;
 273 the large-encoder temporal analysis was specified before its temporal scores were known,
 274 and later scope decisions and added comparisons are identified where they are reported
 275 (Supplementary Notes 16 and 17).

276 4.3 Extrapolation screen

277 The extrapolation screen draws on a fixed vocabulary of power, exponential, linear and
 278 constant forms. On validation data this vocabulary extrapolates better than an eight-form
 279 one that adds quadratic, cubic, square-root and logarithmic terms. Given a point cloud,
 280 a fixed rule takes the candidate fitted regime to be the contiguous stretch in which one
 281 admitted form fits with high R^2 , and the extrapolation region to be the complementary
 282 region. The search runs from both ends and fit quality decides whether the low- or the
 283 high-value direction is retained. The screen reports two statistics. The split-conformal
 284 nonconformity [28] is the mean absolute residual in the extrapolation region divided by the
 285 90th percentile of the absolute residuals in the fitted regime, and the extrapolation-residual
 286 ratio is the median absolute log-residual in the extrapolation region divided by its value
 287 in the fitted regime. Each form is fitted from several initializations, and each denominator
 288 carries a floor, one percent of the median response magnitude in the fitted regime for the
 289 nonconformity and 0.01 for the median log-residual, so that a near-perfect fit cannot produce
 290 an unbounded ratio. Relations with fewer than 20 points are calibrated against size-matched
 291 simulations.

292 4.4 Representation screen

293 The representation screen uses a neural encoder trained only on point clouds from pre-horizon
 294 relations, with no example of a failure and without any anomaly supervision; its learned
 295 score is the distance of a new cloud’s embedding from those of the training relations. At
 296 query time the encoder receives an unordered, normalized (x, y) cloud and nothing else: no
 297 equation, units, date, law name or post-cutoff label.

298 The large encoder, a permutation-invariant set transformer [29] aligned during training
 299 to a network that reads the formulas’ expression trees, is trained on point clouds derived
 300 from 3,441 dimensionless reductions of pre-1900 laws. Held-out calibration and pre-1900
 301 test formulas are separated by law family (Supplementary Fig. 3), and grouping the test
 302 formulas by mechanism gives the nine pre-1900 control groups. The weights used for scoring
 303 are those selected by a rule fixed before any later-physics evaluation. Its learned score
 304 averages four ranks, each transformed by its own pre-1900 calibration empirical cumulative
 305 distribution function: cosine distances to the training formulas’ expression-tree embeddings,
 306 to their means over formulas treated as the same law and to mean point-cloud embeddings
 307 for each formula, and a shrinkage-Mahalanobis distance [30, 31] to the distribution of those
 308 mean point-cloud embeddings. Scores are summarized by the median over clouds within a
 309 formula and then within a law family or control group, and family- and group-level AUROCs
 310 use 2,000 grouped-bootstrap replicates; unless stated otherwise, bracketed intervals in this
 311 paper are 95% bootstrap intervals resampled at that level. The untrained network is scored
 312 by the identical procedure. Architecture, training schedule and corpus sizes are given in
 313 Supplementary Note 16.

314 To build the regenerated controls we regenerated each of the nine pre-1900 control groups
 315 from one representative formula reduction, using the 55-family benchmark’s point count,
 316 x -jitter, repeat count, noise range and heteroscedastic profile, and left the trained weights,
 317 calibration, score and aggregation unchanged. The same-generator set and the 1950-horizon
 318 extension are described in Supplementary Note 16.

319 The compact encoder is trained on 205 pre-1900 point clouds from 40 families, without
 320 unit information. The formula expressions behind these point clouds were drawn from a
 321 machine-assembled formula graph, part of which was generated with language models or
 322 by combinatorial construction. A record was admitted only by exact name match to, or
 323 derivation ancestry from, a law family in a registry of dated pre-1900 sources, which supplies
 324 the family’s date and citation; the expressions themselves were not individually checked
 325 against the original source texts (Supplementary Note 8). No synthetic departure example

enters its training. Leave-family-out calibration holds out each of the 40 training families in full, and the threshold, the 95th percentile of the 40 held-out family scores, is fixed before evaluation. Scores are summarized by the median over eight resampled copies of each dataset and then over the datasets within a family. The generic-function control encoder shares the compact encoder’s architecture and training and is trained on 205 point clouds from smooth generic expressions in 40 pseudo-families matched to the historical set in dataset count, family-size profile and the elementary function types present. Architectures, training schedules and the retraining seeds are given in Supplementary Note 8.

4.5 Post-1900 formula benchmark

The 55-family benchmark contains 55 law families first valid between 1901 and 1996, each carrying a primary historical citation; thermionic emission [32] is one example. Every law is represented by a one-dimensional form $y = f(x)$ in dimensionless variables over a fixed support, sampled eight times at 120 points with varying nuisance parameters, sampling and 0.5–2.5% noise, giving 440 datasets across 55 families. Special reductions, such as the Bardeen–Cooper–Schrieffer interpolation [33], the two-flavour neutrino survival curve [34] and the ideal quantum-Hall staircase [35], are listed in Supplementary Note 15. No benchmark family overlaps a family of the pre-1900 corpus.

The compact encoder’s learned score is summarized by the median over resampled copies, datasets and families and compared with its fixed threshold. The extrapolation score reruns the full form selection and extrapolation procedure against the $n = 120$ thresholds of the size-matched classical calibration (785 simulations per size; below). Eleven cited pre-1900 families excluded from encoder training are the controls, AUROC intervals resample families, and twelve of the post-1900 laws deliberately overlap the affine, power or exponential forms. The eleven additional generated cases are described in Supplementary Note 17.

4.6 Historical measurements and controls

We analyse the published COBE-FIRAS monopole spectrum with its pointwise errors, five Bertozzi values digitized from the 1964 report, NIST specific-heat reference polynomials for six materials, Onnes resistance values including upper limits, Millikan sodium stopping potentials and Solar-System orbital constants. For each case we record the model set, fitted parameters, observation model, fitted regime, adaptive choices, decision rule and output. We evaluate null distributions under the pre-1900 and successor predictions on the same x -grid; explicit error floors stand in for unavailable covariance and for digitization uncertainty, and we vary them in sensitivity analysis.

To test whether the reduction in discrepancy is specific to the successor prediction, five wrong-shape forms, one per case, and five dimensionally coherent wrong-mechanism forms are refitted on the same case grids and observation models with 1,000 parametric-bootstrap replicates. The wrong-mechanism forms substitute fermionic occupation for photon statistics, a proton for the electron rest-energy scale, an Einstein oscillator [36] for Debye acoustic modes [20], Arrhenius transport for the superconducting transition, and Compton recoil for the photoelectric slope.

Sixteen measured series were taken unmodified from the NIST Dataplot and Statistical Reference Dataset collections; their selection, the specification of each series before scoring, and the scoring and grouping procedure are described in Supplementary Notes 14 and 18.

4.7 Synthetic suites, baselines and localization

The baseline comparison uses 36 synthetic datasets, on which the extrapolation screen is compared with its two statistics. Cumulative-sum (CUSUM) change-point detection [3] and Ramsey’s regression specification error test (RESET) [37] are compared with the extrapolation-residual ratio on a separate suite of 16 phenomena (Supplementary Note 3; Supplementary Fig. 2d–f). The mechanism-level benchmark contains 12 classical and 12 synthetic departure families with eight datasets each; its 5% and 1% thresholds come from 785 valid size-matched calibration simulations, and uncertainty is resampled by departure or classical family. The compact encoder is evaluated on 192 datasets over the same 24

378 families. Instrument tests apply calibration drift, sensor saturation, censoring, x -axis error,
379 outliers, correlated noise and heteroscedasticity to classical synthetic relations while leaving
380 the underlying law intact, and boundary intervals are obtained by split-conformal calibration
381 of the normalized localization error separately at $n = 40, 80$ and 160 .

382 5 Data availability

383 The public COBE-FIRAS, Solar-System, Bertozzi, NIST, Millikan and Kamerlingh Onnes
384 inputs are identified in the Supplementary Information. The data package contains source
385 URLs and hashes; the 55-family benchmark and its 440 generated point clouds; the 11
386 pre-1900 control families and their 88 generated datasets; all 128 individual scores of the
387 measured series; the pre-1900 training, calibration and evaluation manifests; and every
388 numerical result used in the figures. The project repository is available at <https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons>, and the release used here
389 will be archived on Zenodo on acceptance under the concept DOI <https://doi.org/10.5281/zenodo.22138620>, which resolves to the most recent version.
391

392 6 Code availability

393 Code is available at <https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons>. The repository contains corpus checks,
394 point-cloud generators, evaluators, statistical calibration and robustness experiments, model
395 definitions, training configurations, figure scripts and the validation scripts. The release
396 tagged `epoch-manuscript-2026-10-04` contains the code and result files used for the
397 numbers reported here.
398

399 7 Acknowledgements

400 We thank NASA LAMBDA, the NIST Cryogenic Material Properties database and the
401 KNAW digital archive for public data access. This research received no specific grant from
402 any funding agency in the public, commercial, or not-for-profit sectors.

403 8 Author contributions

404 C.H. conceived the framework, designed and built EPOCH, ran the experiments, and wrote
405 the manuscript. J.C. contributed to the benchmark construction, case-study analysis, and
406 manuscript revision. Both authors reviewed and approved the final manuscript.

407 9 Competing interests

408 The authors declare no competing interests.

409 **Correspondence and requests for materials** should be addressed to C.H.
410 (ch2067@cam.ac.uk).

References

- [1] Marc C. Kennedy and Anthony O’Hagan. Bayesian calibration of computer models. *Journal of the Royal Statistical Society B*, 63:425–464, 2001. 411
413
- [2] Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning*, 16:494–591, 2023. 414
415
- [3] E. S. Page. Continuous inspection schemes. *Biometrika*, 41:100–115, 1954. 416
- [4] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017. 417
418
419
- [5] Hanchen Wang, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, Payal Chandak, Shengchao Liu, Peter Van Katwyk, Andreea Deac, Anima Anandkumar, Karianne Bergen, Carla P. Gomes, Shirley Ho, Pushmeet Kohli, Joan Lasenby, Jure Leskovec, Tie-Yan Liu, Arjun Manrai, Debora Marks, Bharath Ramsundar, Le Song, Jimeng Sun, Jian Tang, Petar Veličković, Max Welling, Linfeng Zhang, Connor W. Coley, Yoshua Bengio, and Marinka Zitnik. Scientific discovery in the age of artificial intelligence. *Nature*, 620:47–60, 2023. 420
421
422
423
424
425
426
- [6] Keyulu Xu, Mozhi Zhang, Jingling Li, Simon S. Du, Ken-ichi Kawarabayashi, and Stefanie Jegelka. How neural networks extrapolate: from feedforward to graph neural networks. In *International Conference on Learning Representations*, 2021. 427
428
429
- [7] Roger Guimerà, Ignasi Reichardt, Antoni Aguilar-Mogas, Francesco A. Massucci, Manuel Miranda, Jordi Pallarès, and Marta Sales-Pardo. A Bayesian machine scientist to aid in the solution of challenging scientific problems. *Science Advances*, 6:eaav6971, 2020. 430
431
432
- [8] Silviu-Marian Udrescu and Max Tegmark. AI Feynman: a physics-inspired method for symbolic regression. *Science Advances*, 6:eaay2631, 2020. 433
434
- [9] D. J. Fixsen, E. S. Cheng, J. M. Gales, J. C. Mather, R. A. Shafer, and E. L. Wright. The cosmic microwave background spectrum from the full COBE FIRAS data set. *The Astrophysical Journal*, 473:576–587, 1996. 435
436
437
- [10] Lord Rayleigh. Remarks upon the law of complete radiation. *Philosophical Magazine*, 49:539–540, 1900. 438
439
- [11] James H. Jeans. On the partition of energy between matter and aether. *Philosophical Magazine*, 10:91–98, 1905. 440
441
- [12] William Bertozzi. Speed and kinetic energy of relativistic electrons. *American Journal of Physics*, 32:551–555, 1964. 442
443
- [13] Pierre Louis Dulong and Alexis Thérèse Petit. Recherches sur quelques points importants de la théorie de la chaleur. *Annales de Chimie et de Physique*, 10:395–413, 1819. 444
445
- [14] Heike Kamerlingh Onnes. Further experiments with liquid helium. G. On the electrical resistance of pure metals, etc. VI. On the sudden change in the rate at which the resistance of mercury disappears. *Communications from the Physical Laboratory of the University of Leiden*, 124c, 1911. 446
447
448
449
- [15] Paul Drude. Zur elektronentheorie der metalle. *Annalen der Physik*, 306:566–613, 1900. doi: 10.1002/andp.19003060312. 450
451
- [16] R. A. Millikan. A direct photoelectric determination of Planck’s “h”. *Physical Review*, 7:355–388, 1916. 452
453
- [17] Max Planck. Über das gesetz der energieverteilung im normalspectrum. *Annalen der Physik*, 309:553–563, 1901. doi: 10.1002/andp.19013090310. 454
455
- [18] Albert Einstein. Zur elektrodynamik bewegter körper. *Annalen der Physik*, 322:891–921, 1905. doi: 10.1002/andp.19053221004. 456
457

- [19] Albert Einstein. Über einen die erzeugung und verwandlung des liches betreffenden heuristischen gesichtspunkt. *Annalen der Physik*, 322:132–148, 1905. doi: 10.1002/andp.19053220607.
- [20] Peter Debye. Zur theorie der spezifischen wärmen. *Annalen der Physik*, 344:789–839, 1912. doi: 10.1002/andp.19123441404.
- [21] Johannes Kepler. *Harmonices Mundi, Book V*. 1619.
- [22] Robert Boyle. *A Defence of the Doctrine Touching the Spring and Weight of the Air*. Thomas Robinson, Oxford, 1662.
- [23] Raffaele Tito D’Agnolo and Andrea Wulzer. Learning new physics from a machine. *Physical Review D*, 99:015014, 2019.
- [24] Michael Schmidt and Hod Lipson. Distilling free-form natural laws from experimental data. *Science*, 324:81–85, 2009.
- [25] Chenxi He and Jingqiu Chen. A search-free foundation model for symbolic regression of physical laws. *Research Square*, 2026. doi: 10.21203/rs.3.rs-10495716/v1. Preprint, not peer reviewed.
- [26] A. Matthiessen and C. Vogt. On the influence of temperature on the electric conducting-power of alloys. *Philosophical Transactions of the Royal Society of London*, 154:167–200, 1864.
- [27] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57:289–300, 1995.
- [28] Vladimir Vovk, Alexander Gammernan, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- [29] Juho Lee, Yoonho Lee, Jungtaek Kim, Adam R. Kosiorek, Seungjin Choi, and Yee Whye Teh. Set Transformer: a framework for attention-based permutation-invariant neural networks. *Proceedings of Machine Learning Research*, 97:3744–3753, 2019.
- [30] Prasanta Chandra Mahalanobis. On the generalised distance in statistics. *Proceedings of the National Institute of Sciences of India*, 2:49–55, 1936.
- [31] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88:365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.
- [32] Owen W. Richardson. On the negative radiation from hot platinum. *Proceedings of the Cambridge Philosophical Society*, 11:286–295, 1901.
- [33] John Bardeen, Leon N. Cooper, and J. Robert Schrieffer. Theory of superconductivity. *Physical Review*, 108:1175–1204, 1957. doi: 10.1103/PhysRev.108.1175.
- [34] Ziro Maki, Masami Nakagawa, and Shoichi Sakata. Remarks on the unified model of elementary particles. *Progress of Theoretical Physics*, 28:870–880, 1962. doi: 10.1143/PTP.28.870.
- [35] Klaus von Klitzing, Gerhard Dorda, and Michael Pepper. New method for high-accuracy determination of the fine-structure constant based on quantized hall resistance. *Physical Review Letters*, 45:494–497, 1980. doi: 10.1103/PhysRevLett.45.494.
- [36] Albert Einstein. Die plancksche theorie der strahlung und die theorie der spezifischen wärme. *Annalen der Physik*, 327:180–190, 1907. doi: 10.1002/andp.19063270110.
- [37] J. B. Ramsey. Tests for specification errors in classical linear least-squares regression analysis. *Journal of the Royal Statistical Society B*, 31:350–371, 1969.